English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Applied Mathematics / Mathematics, 19 April 2026 Test 1 _____ *The exact grading formula will depend on the results. As an approximation, use 2 + the number of points. Every answer must be justified and supported by clear, detailed calculations. Electronic devices (except standard calculators) and additional materials are prohibited. Time allowed: 4 hours. Good luck!*

Problem 1. (1.25 points)

1. (0.25 points) An urn contains n balls, and all possible numbers of white balls are equally likely. We add one white ball. What is the probability that a randomly chosen ball is white?

2. (0.25 points) An urn contains n white and m black balls. Two players draw in succession and *then* return the drawn balls. The first to draw a white ball wins. What is the probability that the first player wins? What happens as $m \to \infty$?

3. (0.25 points) We have two urns, 5 white balls and 5 black balls. How should we distribute the balls between the urns, by number and colour, to maximize the probability of drawing a white ball after choosing an urn at random and then a ball at random from it?

4. (0.5 points) Ivan has only 10-stotinka coins. The price of each item in a shop, in stotinki, ends in a uniformly random digit in $\{0, 1, \ldots, 9\}$; the final digits are independent. Ivan buys $n \geq 1$ items. What is the probability that he can pay exactly, without receiving change[1]? What is the expected number of items he must take before he can pay exactly for the first time[2]?

Problem 2. (1.5 points)

1. (0.25 points) Let $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B|A) = 0.4$ and $\mathbb{P}(A \cup B) = 0.8$. Find $\mathbb{P}(B)$ and determine whether A and B are independent.

2. (0.25 points) A discrete random variable X has probability generating function $g_X(s) = c(2+s)^4$. Find c, $\mathbb{E}[X]$ and $\mathbb{P}(X = 2)$.

3. (0.25 points) Let $X \sim \text{Poi}(\lambda_1)$ and $Y \sim \text{Poi}(\lambda_2)$ be independent. Define $T = X + Y$. Find the conditional distribution and expectation of X given T.

4. (0.25 points) Let $X_1, X_2, X_3 \overset{iid}{\sim} \text{Geo}(p)$ and $Y = X_1 + X_2 + X_3$. Calculate $\mathbb{P}(X_1 > X_2 + X_3)$ and $\mathbb{P}(X_1 > X_2 + X_3 \mid Y = 15)$.

5. (0.5 points) A factory produces items that are defective with probability 0.02, independently. You inspect items until you find 15 defective ones. Estimate the probability that you need to inspect at most 500 items.

Problem 3. (1 point) Let $X_1, \ldots, X_n$ be independent, identically distributed discrete random variables and $S_n = X_1 + \cdots + X_n$.

1. (0.25 points) Find $\mathbb{E}[X_i \mid S_n]$, $1 \leq i \leq n$.[3]

2. (0.25 points) Prove that if f is strictly monotone and X, Y are random variables such that $f(Y)$, $\mathbb{E}[X \mid Y]$ and $\mathbb{E}[X \mid f(Y)]$ are well defined, then $\mathbb{E}[X \mid f(Y)] = \mathbb{E}[X \mid Y]$.

Let $f : I \to \mathbb{R}$, where I is an interval. If for every $a, b \in I$ and $\lambda \in [0,1]$, $f(\lambda a + (1-\lambda)b) \leq \lambda f(a) + (1-\lambda) f(b)$, then f is called convex.[4] Convex functions satisfy *Jensen's inequality*: $f(\mathbb{E}X) \leq \mathbb{E}f(X)$ for every random variable X taking values in I.

3. (0.5 points) Let $P_n = X_1 \cdots X_n$. Using the preceding results and Jensen's inequality appropriately, prove that $\mathbb{E}[X_i \mid P_n] \geq \sqrt[n]{P_n}$ for $1 \leq i \leq n$. Here assume $X_1 > 0$.

Problem 4. (1.2 points) A flamingo jumps along the integers as follows. Starting at 0, at each step it chooses uniformly one of the next 3 integers and jumps to it. Thus, from m it moves to $m+1, m+2$ or $m+3$, with equal probabilities. We are interested in its possible paths.

1. (0.2 points) Model the experiment: how would you choose $(\Omega, \mathcal{F})$?

[1] That is, his bill in stotinki is a multiple of 10.
[2] That is, the number of items until their total cost in stotinki first becomes a multiple of 10.
[3] Throughout this problem, assume that the conditional expectations are well defined.
[4] If f is twice differentiable, this condition is equivalent to $f''(x) \geq 0$ for $x \in I$.

2. (0.25 points) Let S_n be its position after n steps, with $S_0 = 0$. Find $\mathbb{E}[S_n]$, $\mathrm{Var}(S_n)$ and $\rho_{S_n,S_{n+k}}$. What happens to the correlation as $k \to \infty$?

Write $\tau_n : \Omega \to \mathbb{N} \cup \{\infty\}$ for the number of moves until the flamingo reaches n, if it does, and set $\tau_n = \infty$ otherwise.

3. (0.25 points) Calculate $\mathbb{E}[\tau_4]$ and $\mathbb{E}[\tau_4 \mid \tau_4 < \infty]$.

4. (0.5 points) Prove that
$$\mathbb{P}(\tau_n < \infty) = \frac{1}{2}\left(1 + 3^{-n/2} T_n\left(-\frac{1}{\sqrt{3}}\right)\right),$$
where T_n is the nth Chebyshev polynomial of the first kind.[5]

Problem 5. *(Bonus)* (1 point) Let X take finitely many values in $(0, \infty)$. What distribution of X maximizes/minimizes $\mathbb{E}[X]\mathbb{E}[X^{-1}]$? What is the corresponding maximum/minimum?

- *Hint:* You may use
$$\max\left\{\sum_{i=1}^{n} p_i^2 \;\middle|\; p_i \geq 0, \sum_{i=1}^{n} p_i = 1\right\} = \max\left\{\sum_{i=1}^{n-1} p_i^2 \;\middle|\; p_i \geq 0, \sum_{i=1}^{n-1} p_i = 1\right\}.$$

[5] Recall that $T_n(\theta) = \cos(n \arccos\theta)$ for $-1 \leq \theta \leq 1$.